IMP PROJECT RESEARCH DEFENSE

Evidence before leaderboard claims

A repaired evidence contract turns a contaminated benchmark and a failed hypothesis into defensible research.

MiT-B3 U-Net versus nnU-Net v2

Negative contour-channel ablation
Protected test remains sealed; this is not state of the art

EVIDENCE ATLAS

research report | July 2026

WHY THE STUDY NEEDED REPAIR

Leakage changed the meaning of every favorable score

Three patient identities and 13 rows crossed legacy split boundaries, so the old leaderboard became historical evidence rather than a scientific ranking.

3

crossing patient IDs

13

cross-boundary rows

- Patient overlap violates independent patient-level interpretation.
- Old test-v2 values remain visible only as contamination-aware history.
- The repair starts with identity groups, not model tuning.

Two hypotheses, two evidence classes

> **The architecture comparison and the contour-channel ablation answer different questions and must not be collapsed into one ranking.**

RQ1

How do complete MiT-B3 and nnU-Net systems trade overlap, boundary quality, precision, and recall?

System comparison
Single recorded run
Descriptive inference

RQ2

Does a saliency-constrained contour channel improve a matched MiT-B3 control?

Matched intervention
Three selected seeds
Conditional paired inference

CLICK A MODULE TO INSPECT IT

The claim can only be as strong as its weakest evidence boundary

> **Every model output passes through data, protocol, metric, and claim gates before it reaches the paper or demo.**

01
Data Audit

02
Preprocessing

03
Models

04
Robust Evaluation

05
Loop206 Ablation

06
Evidence-bound Demo

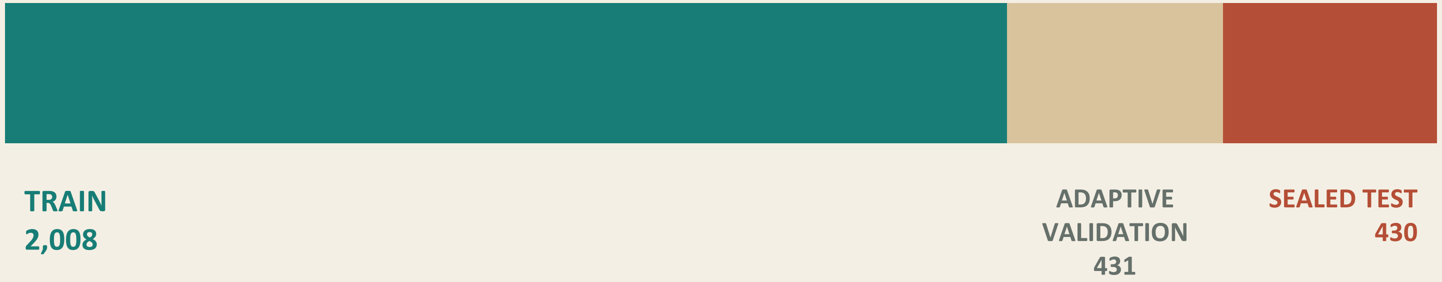
Interactive navigation and smooth transitions are available in the offline HTML edition.

Clean-v3 repairs identity leakage before model comparison

> **The leakage-audited manifest contains 2,869 images with no detected cross-split identity-group overlap.**

2,869

leakage-audited images



IDENTITY / EXACT DUPLICATE / PERCEPTUAL DUPLICATE / SPLIT GROUP

Validation was opened adaptively. The 430-row test partition remains sealed.

The comparison is between complete systems, not isolated encoders

> **MiT-B3 uses a fixed preprocessing-aware 384x384 pipeline; nnU-Net v2 uses raw RGB and a self-configuring 256x256 system.**

MiT-B3 U-Net

384 × 384
LAB-luminance CLAHE
Percentile stretch + median filter
Fixed preprocessing-aware system

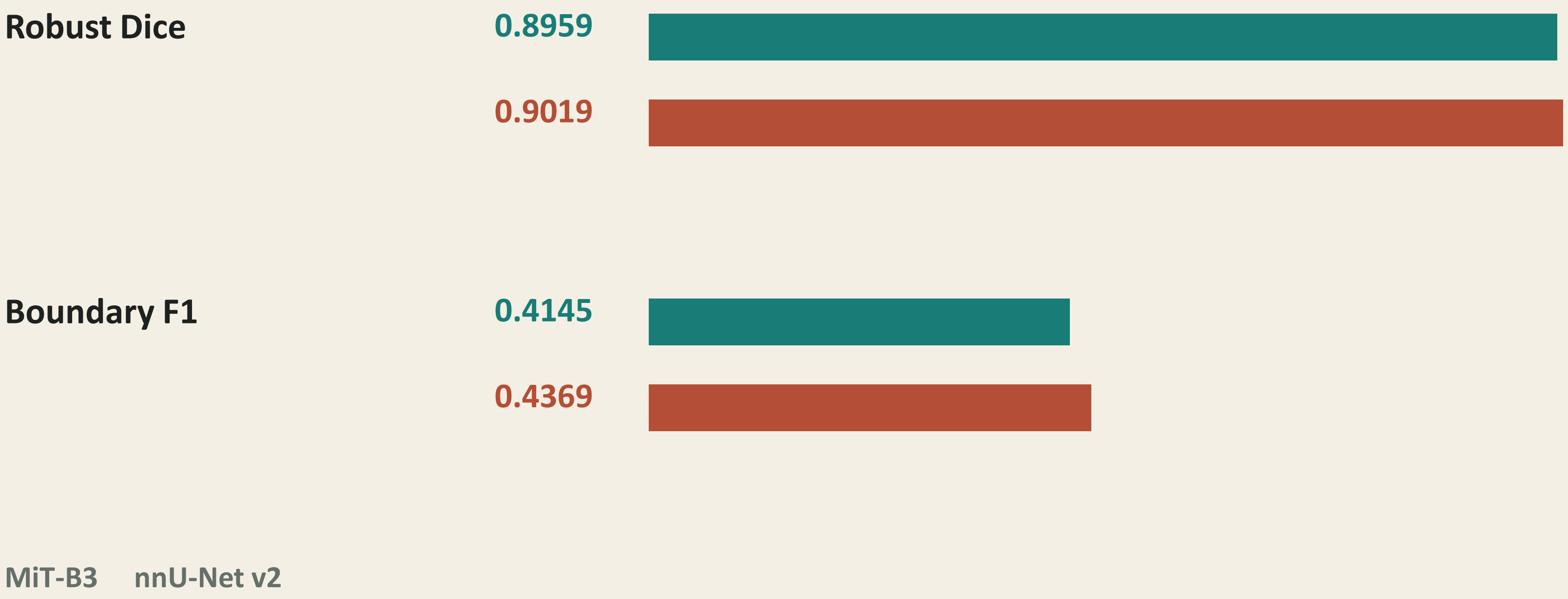
nnU-Net v2

256 × 256
Raw RGB
Generated plans
Self-configuring 2D system

Different resolution, decoder, loss, augmentation, and policy: this is not an isolated encoder ablation.

nnU-Net records higher point estimates, not statistical superiority

On adaptive development-validation, nnU-Net v2 has higher robust Dice and boundary F1 point estimates; MiT-B3 retains slightly higher precision.

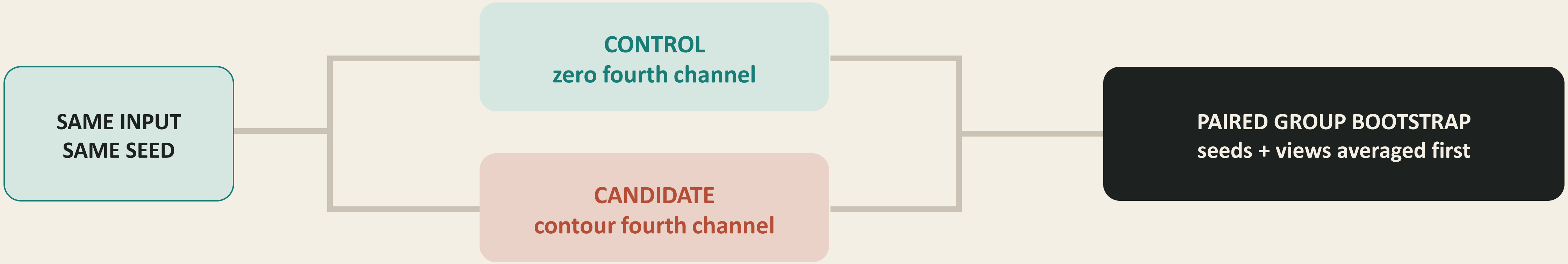


Point estimates only
Single recorded run
No paired confidence interval

MiT-B3 retains slightly higher precision; metric trade-offs remain material.

Loop206 isolates one intervention: the fourth input channel

Control and candidate share architecture, preprocessing, seeds, budget, and loss; only the zero versus contour channel changes.



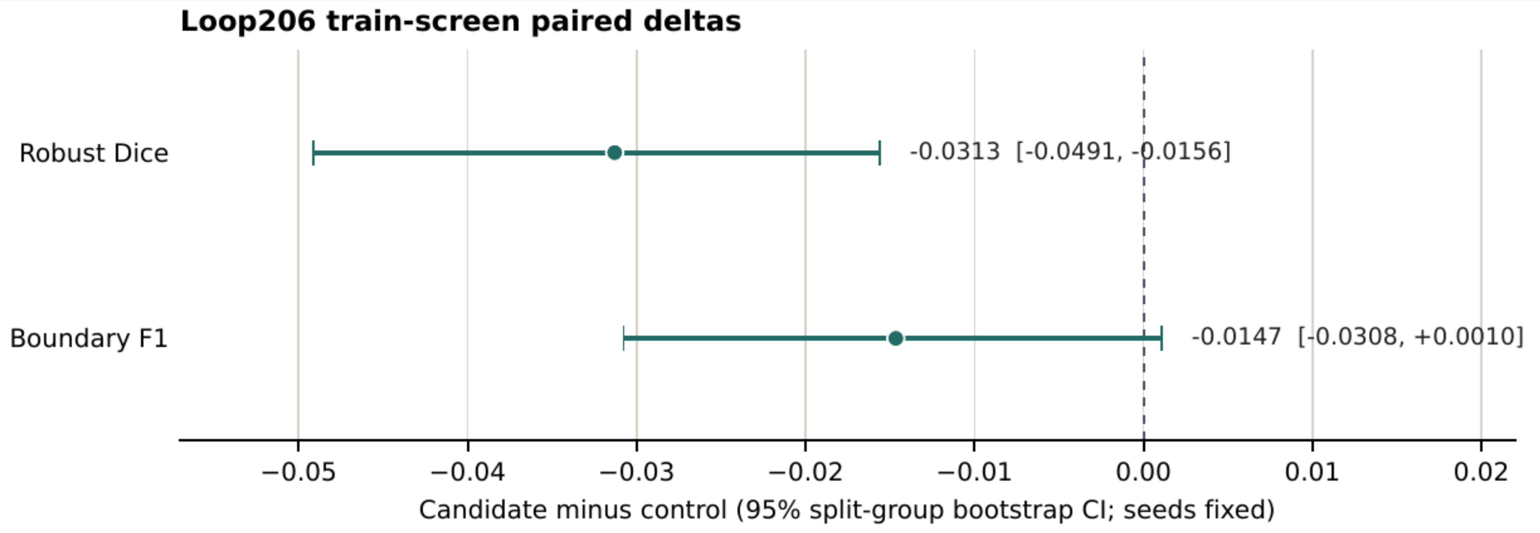
308
fit groups

76
holdout groups

3
paired seeds

The contour channel fails the primary gate

Seed-and-view-averaged robust Dice changes by -0.0313; its conditional 95% interval lies entirely below zero.



REJECT

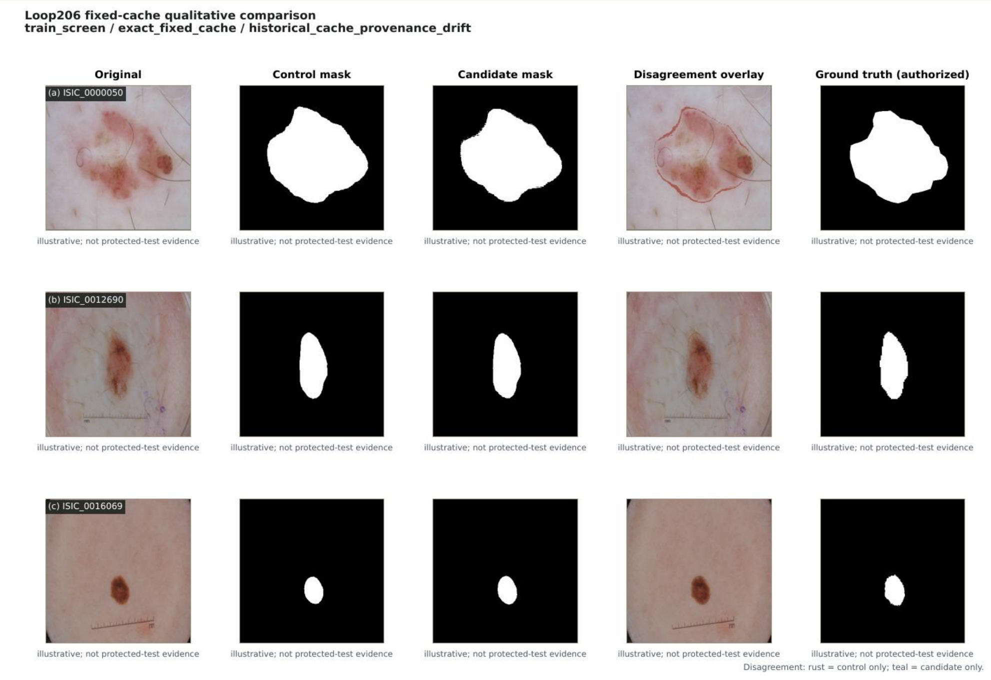
Dice Δ -0.0313
95% CI [-0.0491, -0.0156]

BF1 Δ -0.0147
95% CI [-0.0308, 0.0010]

Candidate rejected before protected evaluation.

The demo compares only what the registry authorizes

The public workbench supports exact fixed-cache comparison; arbitrary uploads remain control-only because candidate prior parity is 0/76.



0/76

candidate prior parity

FIXED CACHE
control + candidate + authorized truth

ARBITRARY UPLOAD
control only

Non-clinical research demo; not a diagnosis.

Claims are reproducible; full retraining is not yet clone-runnable

> The release binds claims, tables, figures, PDF, model IDs, and source reports by hash while explicitly disclosing missing experiment artifacts.

HASH-BOUND

Claims
Tables + figures
Compiled PDF
Model IDs
Source reports

RUNTIME-BOUND

Checkpoints
Data
Caches
Secrets
Startup verification

NOT YET CLONE-RUNNABLE

Loop191/192 configs
Implementation modules
Paired predictions
Physical laptop evidence

Protected test remains sealed.

DEFENSIBLE TAKEAWAY

A narrower conclusion is the stronger scientific result

nnU-Net records higher development-validation point estimates, the hard contour prior fails its matched train-screen gate, and no protected-test or state-of-the-art claim is authorized.

01 Repair leakage before optimizing models.

02 Use strong baselines without hiding contract differences.

03 Publish negative ablations when they reject a mechanism.

NEXT DEFENSIBLE EXPERIMENT Reproduce both systems under one geometry contract before any protected test.